SEQUENCE 15

Consolidation and transition to Year 12

25 May–11 June 2027

15.1 Review of calculation, functions and geometry

Example 1 · Annual review – part A.

1. Reduce $168/252$ to lowest terms; calculate $2^{-3}+1/4$.
2. Factorise x^2-6x+9, then solve $(x-3)^2=4$.
3. Solve $\dfrac{x-2}{x+1}\geq 0$.
4. For $f(x)=4-(x-1)^2$ on $[-2,3]$, give the maximum and the abscissae where $f(x)=0$.
5. With $A(0,1)$, $B(4,3)$, $C(6,4)$, prove collinearity and give an equation of the line.

Solution. 1. $168/252=2/3$, and $2^{-3}+1/4=1/8+2/8=3/8$. 2. $(x-3)^2$; $x-3=\pm 2$, so $S=\{1,5\}$. 3. Forbidden -1, zero 2; signs $+,-,+$, so $S=(-\infty,-1)\cup[2,+\infty)$. 4. The square is positive or zero, so $f(x)\leq 4$, with equality at $x=1$. The zeros are $x=-1,3$, both in the domain. 5. $\overrightarrow{AB}=(4,2)$, $\overrightarrow{AC}=(6,3)=\frac{3}{2}\overrightarrow{AB}$. An equation is $x-2y+2=0$.

Teacher note. Indicative marking: 10 points, 2 per question. For question 3, reserve one point for the domain and the endpoints. For question 5, ask for the conclusion after the calculation.

15.2 Review of data, probability and modelling

Example 2 · Annual review – part B.

1. Calculate the mean and median of $2;4;4;6;9$. Determine Q_1, Q_3.
2. A price drops by 15%, then rises by 20%. Give the overall rate and the rate that then restores the initial price.
3. In a fictional population of 100 people, 40 play sport (S), 30 come by bike (V), 18 do both. Construct the two-way table and calculate $P_S(V)$, $P_V(S)$ for an equally likely draw.
4. For a rectangle of perimeter 24 m, express the area as a function of one dimension x and prove its maximum.

Solution. 1. Mean $25/5 = 5$, median 4. Quartile ranks 2 and 4, so $Q_1 = 4$, $Q_3 = 6$.
2. Overall multiplier $0.85 \times 1.20 = 1.02$: an increase of 2%. Reverse rate $1/1.02 - 1 = -1/51 \approx -1.96\%$.
3. Table: $S \cap V = 18$, $S \cap \overline{V} = 22$, $\overline{S} \cap V = 12$, $\overline{S} \cap \overline{V} = 48$. $P_S(V) = 18/40 = 45\%$, $P_V(S) = 18/30 = 60\%$.
4. $A(x) = x(12 - x) = 36 - (x - 6)^2$ on $(0, 12)$. The maximum is 36 m^2, attained for the square of side 6 m.

Teacher note. Indicative marking: 10 points, respectively 2, 2, 3, 3. The review of parts A and B can be offered in one hour, then revisited with the course materials.

15.3 Choosing a strategy and preparing what comes next

Example 3 · A program to explain. For a given function f, explain the following program. What does it return with $f(x) = x^2$?

```
def tableau(f):
    valeurs = []
    for x in range(-3, 4):
        valeurs.append((x, f(x)))
    return valeurs
```

Solution. It returns a list of abscissa–image pairs for the integers from -3 to 3 inclusive: $(-3,9), (-2,4), (-1,1), (0,0), (1,1), (2,4), (3,9)$. The bound 4 of `range` is excluded. The table does not describe the images of all real numbers between -3 and 3.

Key point · Six skills to mobilise. **Search**: try things out, organise leads. **Model**: choose variables, domain and assumptions. **Represent**: move from text to table, figure or graph. **Reason**: justify and distinguish conjecture from proof. **Calculate**: choose a form and check the operations. **Communicate**: explain clearly with suitable notation and units.

Example 4 · End-of-year portfolio. Choose a successful piece of work and a corrected one. For each, identify the skills used, explain one method and state a personal goal for the following year. Then redo an exercise matching this goal without looking at the correction.

Solution. There is no single numerical answer. A satisfactory piece of work precisely identifies the problem, the chosen method, a justification and a check. The goal must be observable, for example: “state the forbidden values before solving a quotient equation”, rather than “get better at calculation”.

Teacher note. To present the Year 12 pathways, rely on the courses actually offered in the school and their programmes in force. The booklet sets no automatic mark threshold for guidance.